is the presence of p+ layer that is added to the drain side. This p+ layer is also called injecting layer. The next layer is n+ layer also called as buffer layer. There is a p-n junction J_1 between the injecting layer and the buffer layer. There are two more p-n junctions J_2 and J_3 as shown in Figure 6.23. The junction J_1 blocks reverse voltage. The junction J_2 blocks forward voltage when IGBT is off.

Working Principle: When a positive voltage is applied between the gate and source, the power MOSFET turns 'ON' and acts as a low resistance between the base and collector of the PNP transistor, thereby the IGBT is turned 'ON'. When there is no gate to source voltage the MOSFET is turned off and hence the PNP transistor is also off because no longer base current is supplied. Thus, the IGBT acts as a switch.

Application of IGBT

1. The IGBT is used in medium to high power application like SMPS, traction motor control etc.
2. Large IGBT modules consist of many devices in parallel have the capability to control current in hundreds of amperes with blocking voltage of 6500 V.

6.15 INTEGRATED CIRCUIT (IC)

An integrated circuit consists of number of circuit components (e.g., transistors, diodes, resistors, capacitors etc.,) and their inter connections in a single small package to perform a complete electron function. Those components are formed and connected within a small chip of semiconductor material. The following points are worth noting about Integrated Circuits.

i. In an IC, the various components are automatically part of a small semiconductor chip and the individual components cannot be removed or replaced. This is in contrast to discrete assembly in which individual components can be removed or replaced if necessary.

ii. The size of an IC is extremely small. In fact, ICs are so small that it normally need a microscope to see the connections between the components. Figure 6.24 shows a typical semiconductor chip.

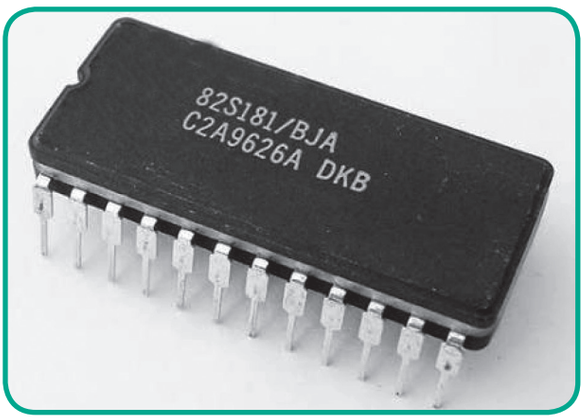


Figure 6.24 Integrated Circuit (IC)

iii. No components of an IC are seen to project above the surface of the chip. This is because all the components are formed within the chip.

ELEC_EQUIP_Chapter06.indd 143 11/10/2020 11:44:10 AM